

Mateo Altomare

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EDUCATION

University of South Carolina
B.S. Mechanical Engineering

GPA: 3.91
Aug 2024 – May 2028

River Bluff High School
College Preparatory Diploma, A Honor Roll

GPA: 4.9/5.0
Aug 2020 – May 2024

RESEARCH EXPERIENCE

Integrated Multiphysics and Systems Engineering Laboratory (iMSEL)

Columbia, SC

Undergraduate Research Assistant, Dr. Yi Wang

June 2025 – Present

- Co-authoring paper for ASME IDETC 2026 presenting depth-aware linear affine transformation method for pixel-level IR-to-RGB image registration in maritime sensing applications
- Deployed real-time YOLO object detection on embedded Raspberry Pi platform for automated target identification and tracking in marine radar PPI displays
- Engineered 4-channel multi-spectral detection pipeline fusing thermal IR and RGB imagery with scene-based keypoint extraction for cross-modal sensor calibration
- Trained Detection Transformer (DETR) model on maritime dataset using transfer learning via RGB feature extraction from pre-trained backbones
- Curated multi-modal training datasets through systematic annotation of radar returns, thermal infrared, and visible spectrum imagery for supervised learning pipelines

LEADERSHIP EXPERIENCE

Gamecock Motorsports Formula SAE

Columbia, SC

Chassis Team Co-Lead

August 2024 – Present

- Designed complete chassis architecture from ground up, optimizing for cost, weight, and weldability while ensuring full compliance with FSAE competition rulebook
- Defined subsystem interface specifications and interaction points enabling seamless integration between chassis, suspension, powertrain, and aerodynamics teams
- Developed and delivered CAD training sessions for new team members covering component modeling workflows and design-for-manufacturing principles
- Created visual renderings and technical documentation for fundraising presentations to secure sponsorship funding for vehicle development

American Society of Mechanical Engineers (ASME)

Columbia, SC

Member

August 2024 – Present

- Contributing to team projects, workshops, and technical activities to strengthen mechanical engineering skills
- Participating in industry networking events for professional development and career exploration
- Collaborated on design challenge projects applying engineering principles to develop creative mechanical solutions

ACTIVITIES

HOSA (Health Occupations Students of America)

2020 – 2024

DECA Business Organization

2020 – 2024

Varsity Soccer & Track

2020 – 2024

SKILLS

Engineering Software:

SolidWorks, Creo, Fusion 360, Cura Slicer, Microsoft Office

Programming:

Python, C++, Rust, MATLAB, PyTorch, OpenCV

Hardware:

Raspberry Pi, NVIDIA Jetson Orin